

RECHIDI Ahmed Abdelaaziz

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📍 Sidi Bel Abbes, Algeria

Who I Am

I am a Software Engineer and Cybersecurity student who loves building things that are both fast and secure. Whether it's crafting a smooth Flutter app or engineering a robust backend in NestJS, I focus on writing code that solves real problems for real people. I thrive in the space between functionality and security, constantly looking for ways to make applications more resilient without sacrificing the user experience.

Toolbox & Expertise

Frontend: Flutter (BLoC/Cubit), Dart, Clean Architecture, Retrofit/Dio, Freezed, Equatable, Lottie, FL Chart
Backend: NestJS, Node.js, Express, MongoDB/Mongoose, SQL, Identity (OAuth2.0, JWT Architecture, RBAC)
Infrastructure: Docker, Docker Compose, Firebase, Cloudinary, CI/CD, Linux
Security: OWASP Top 10, Network Security, Ethical Hacking, Data Sanitization, Authentication Systems, Authorization, MFA, Secure Storage

What I've Built

FENNEQCM Mobile App | 📱 *Live (+1k Downloads)*

- **Enterprise Architecture:** Engineered a highly scalable Flutter codebase using feature-based **Clean Architecture** and **BLoC**, decoupling business logic across 15+ modules (Exams, Payments, Analytics) for 100% independent team scalability.
- **Resilient Networking:** Built a robust **Dio Interceptor** ecosystem featuring automated background JWT token refreshing and recursive request retries, resulting in a zero-session-interruption user experience.
- **Interactive UX:** Implemented **flutter_quill** for rich-text medical question authoring and custom **FL Chart** dashboards to visualize real-time performance analytics (Global vs. Specialty progress).
- **Security:** Integrated **Flutter Secure Storage** for encrypted credential management and established a multi-environment build pipeline (Dev/Prod flavors) using Kotlin DSL and Git-excluded signing keys.

FENNEQCM Backend Ecosystem | *NestJS, MongoDB, Docker*

- **Monorepo Scalability:** Architected a **NestJS Monorepo** that synchronized the Student API and Admin Content Manager on a single core engine, reducing maintenance overhead by 40% via shared **libs/common** schemas.
- **Security Architecture:** Engineered a **Zero-Trust** identity system featuring **Google OAuth 2.0**, **JWT Access/Refresh rotation**, and strict RBAC decorators to gate 100+ medical specialties and subscription-locked content.
- **Data Integrity:** Hardened the persistence layer with custom **Mongoose indexing** for clinical case hierarchies and developed specialized validation pipes to sanitize all inputs against NoSQL Injection and XSS attacks.
- **Cloud Infrastructure:** Orchestrated **Cloudinary** for optimized medical diagram delivery and automated transactional workflows using **Nodemailer with Google OAuth2** to ensure 99.9% email deliverability.

E-Commerce Core API | 📄 *Source Code*

- **Generic Design:** Developed a high-performance **Generic Repository Pattern** that abstracted complex Mongoose operations, standardizing high-speed pagination, advanced filtering, and global API response formatting.
- **Identity & Safety:** Built a dual-channel **MFA system** using **Twilio (SMS)** and Nodemailer, ensuring secure account verification before transaction access, coupled with one-way BCrypt password hashing.
- **Business Intelligence:** Engineered a flexible discount engine and real-time inventory management logic to prevent stock race conditions during high-concurrency purchase phases.

Education

ESI Sidi Bel Abbes

Engineering Degree in Computer Science

Expected Sept 2027

Sidi Bel Abbes, Algeria

- Deepening expertise in Software Engineering and Application Security. Actively researching performance optimization in mobile environments and implementing OWASP-aligned security controls.